How Far Can a Routine Blood Count Go? Realistic Performance, Resampling Leakage and External Validation of CBC-Based Dengue Detection in Bangladesh

Tashnimul Hasan Shifat
Dept. of CSE
University of Asia Pacific
Dhaka, Bangladesh
23101142@uap-bd.edu

Sultan Ruhul Kuddus Nayan
Dept. of CSE
University of Asia Pacific
Dhaka, Bangladesh
23101159@uap-bd.edu

Anjuma Alam
Dept. of CSE
University of Asia Pacific
Dhaka, Bangladesh
23101164@uap-bd.edu

Fairoj Salim
Dept. of CSE
University of Asia Pacific
Dhaka, Bangladesh
23101165@uap-bd.edu

Abstract*—Bangladesh recorded more than 321,000 dengue cases in 2023, yet confirmatory tests are often unavailable outside large hospitals, while a complete blood count (CBC) is ordered for almost every febrile patient. We ask how much dengue information routine CBC data really carry, how much class resampling before the train/test split inflates reported results, and whether models transfer to another hospital. On 1,511 patients from Jamalpur (2024), every strong model reached the same leakage-free ceiling: over ten repeated splits that re-ran the whole pipeline, ROC-AUC was 0.688 ± 0.030 for our best pipeline, 0.685 ± 0.032 for Random Forest and 0.683 ± 0.030 for raw-CBC XGBoost. Emulating resampling before the split raised accuracy from about 75% to 92% and ROC-AUC from 0.67 to 0.96 on the same data. As an interpretable component we use the Dengue Clinical Hematological Signature Score (D-CHSS), which converts CBC values into capped, sex-aware deviations from fixed reference ranges grouped into thrombocytopenia, leukocyte and hemoconcentration components; its four features matched the sixteen raw values internally. On an independent cohort from two Dhaka hospitals (2023, $n = 312$), using the ten shared variables, raw-value models trained in Jamalpur transferred poorly (ROC-AUC 0.31–0.69), whereas D-CHSS features reached 0.89 (95% CI 0.85–0.92). Yet the platelet count alone reached 0.95 in Dhaka but only 0.53 in Jamalpur, so CBC-based performance depends strongly on case mix. At 20% prevalence, the Jamalpur model's positive and negative predictive values would be 27% and 97%. A calibrated logistic regression is deployed as a Bengali web application.

***Index Terms*—dengue, complete blood count, external validation, data leakage, explainable AI, XGBoost, SHAP, clinical decision support, Sustainable Development Goals**

I. Introduction

Dengue is caused by four serotypes (DENV-1 to DENV-4) and is spread mainly by *Aedes* mosquitoes [1], [2]. An estimated 390 million infections occur each year, about 96 million of them symptomatic [3]. There is no specific antiviral drug, so early recognition and careful fluid management are what save lives [1], [2]. In Bangladesh, a review of studies from 2000 to early 2024 counted 565,438 cases and 2,587 deaths, and the 2023 season (321,179 cases) was the worst on record [4], [5].

Confirmatory tests are frequently delayed or missing at district and upazila hospitals during outbreaks [2], [6]. A CBC, by contrast, is cheap, automated and almost always performed for a febrile patient, and the textbook dengue picture of falling platelets, low white cells and rising hematocrit is visible in it [1], [7]. Machine learning (ML) on CBC data has shown promise in Bangladesh [8]–[10], but existing models usually feed raw values to a classifier, report accuracy alone, sometimes let test information leak into preprocessing [11], and rarely explain predictions in clinical terms that a pathologist can check [12].

Rather than proposing another classifier, this paper asks three questions: what is the realistic, leakage-free ceiling of CBC-only dengue detection, how much does resampling before the split inflate it, and does a model built in one hospital transfer to another? Its contributions are: (i) a leakage-safe and uncertainty-aware evaluation, with per-model tuned thresholds, bootstrap confidence intervals, DeLong tests and ten repeated splits that re-run the whole pipeline, which establishes the realistic ceiling on the Jamalpur cohort; (ii) a controlled experiment that quantifies how class resampling before the train/test split inflates accuracy and ROC-AUC on the same data; (iii) an external validation on an independent cohort from two Dhaka hospitals, which shows that performance depends strongly on case mix and that features anchored to fixed clinical reference ranges transfer far better than raw values; (iv) D-CHSS, an interpretable component that summarizes thrombocytopenia, leukocyte response and hemoconcentration as reference-range deviations, reported together with its unexpected inverse relationship in the development cohort; and (v) a Bengali-language web application whose model was evaluated on the same test set, with results translated into predictive values at realistic prevalences.

II. Related Work

Population-level forecasting. Dey *et al.* [13] combined hospital admissions with weather data for eleven districts (SVR,

TABLE I
PATIENT-LEVEL DENGUE STUDIES MOST RELEVANT TO THIS WORK

Study	Data	Method	Metric	Value	Main gap
Ho [17]	4,894 ED pts, Taiwan	DNN, DT, LR	AUC	0.84–0.86	needs temperature
Riya [8]	320 CBC, Dhaka	Stacking	Acc.	96.88%	small data
Haque [6]	Hematology	GA, GAN, DT	Acc.	99.49%	synthetic balancing
Tuhin [9]	1,523 CBC, Jamalpur	Stacking, SHAP	Acc.	96.38%	resampling
Hasan [10]	1,523 CBC, Jamalpur	RF, XGB, SVM	AUC	0.69	little interpretation
This work	1,511 CBC, Jamalpur	D-CHSS + XGB	AUC	0.685	single site

Acc. = accuracy; AUC = ROC-AUC. The two are not comparable: on this cohort a model that calls every patient positive already reaches about 68% accuracy.

75% accuracy), Liu *et al.* [14] found XGBoost the most accurate monthly forecaster, and Shiddik *et al.* [5] built a 64-district early warning system with SHAP and Bayesian models. These models plan outbreak responses but cannot diagnose individual patients.

Clinical predictors. A systematic review found that most published rules used platelet count, white-cell count, rash and liver markers, but few were validated [7]. Daumas *et al.* [15] reached 81% sensitivity and 71% specificity with conjunctival redness and leukocyte count, and the multi-country IDAMS study of 7,428 febrile patients confirmed platelet and white-cell counts as the strongest laboratory markers [16].

ML on routine data. Ho *et al.* [17] reached an AUC of about 0.84–0.86 in Taiwan with age, body temperature, WBC and platelet count. Qaiser *et al.* [18] found SVM best on routine laboratory data in Pakistan. In Bangladesh, Riya *et al.* [8] reported 96.88% accuracy on 320 CBC reports, and Haque *et al.* [6] reported 99.49% with GAN-based balancing. Two studies used the same Jamalpur cohort as this work: Tuhin *et al.* [9] reported 96.38% accuracy with a stacking ensemble after resampling, and Hasan *et al.* [10] reported 84.5% accuracy and an AUC of 0.69 with Random Forest. Both used SMOTE-type resampling [19]; the order of resampling and splitting matters, because resampling must be confined to training folds to avoid optimistic estimates [11]. Table I summarizes the most relevant studies. None of them represents the three core blood changes of dengue as explicit, interpretable components.

III. MATERIALS AND METHODS

A. Dataset and Preprocessing

We used the public Dengue Fever Hematological Dataset [20]: 1,523 anonymized records from Jamalpur 250-Bedded General Hospital (February–September 2024), each confirmed by NS1, IgM/IgG or PCR [10]. Records contain gender, age (5–78 years), sixteen CBC parameters and the result, with no missing values. After removing 12 exact duplicates, 1,511 records remained (68.4% positive). A model labelling everyone positive would already reach about 68% accuracy, so accuracy alone is a weak yardstick [21]. Outliers were kept because extreme values such as very low platelets are often the signal itself. The data were split 80/20 with stratification and a fixed seed (42), giving 1,208 training and 303 test records. Imputers and scalers were fitted on the training set only. Fig. 1 shows the full pipeline.

TABLE II
TRAINING-SET CHARACTERISTICS BY CLASS: MEDIAN [IQR] AND MANN–WHITNEY p

Parameter	Negative ($n = 381$)	Positive ($n = 827$)	p
Age (years)	36.0 [26.0, 50.0]	40.0 [28.0, 54.0]	<0.001
Hemoglobin (g/dl)	14.8 [13.3, 15.8]	14.5 [13.1, 15.7]	0.326
Neutrophils (%)	44.0 [41.0, 47.0]	45.0 [43.0, 48.0]	<0.001
Lymphocytes (%)	46.0 [41.0, 48.0]	44.0 [41.0, 47.0]	<0.001
Monocytes (%)	3.0 [3.0, 4.0]	3.0 [2.0, 4.0]	<0.001
Eosinophils (%)	3.0 [2.0, 4.0]	3.0 [2.0, 4.0]	<0.001
RBC (10^6/µl)	5.0 [4.0, 5.0]	5.0 [4.0, 5.0]	<0.001
HCT (%)	46.0 [43.5, 48.0]	45.1 [42.0, 47.7]	0.008
MCV (fl)	90.0 [86.0, 95.0]	89.4 [85.0, 94.0]	0.018
MCH (pg)	30.1 [29.0, 31.3]	29.9 [28.3, 31.4]	0.092
MCHC (g/dl)	31.7 [30.8, 32.8]	32.1 [30.8, 33.5]	0.002
RDW-CV (%)	14.3 [12.9, 15.3]	13.5 [12.3, 14.5]	<0.001
Platelet (10^3/µl)	175.6 [138.0, 221.1]	162.2 [109.4, 233.9]	0.027
MPV (fl)	10.0 [9.0, 10.4]	9.6 [8.8, 10.2]	<0.001
PDW (%)	15.4 [14.3, 16.4]	15.2 [14.1, 16.5]	0.567
PCT (%)	0.14 [0.10, 0.18]	0.14 [0.11, 0.17]	0.095
WBC (10^3/µl)	5.5 [4.8, 6.4]	5.4 [4.6, 6.1]	<0.001

External cohort. For external validation we used the public CBC dataset of Riya *et al.* [8]: 320 suspected dengue patients from two hospitals in Dhaka (May–October 2023), a different city, year and outbreak from the development cohort. It records gender, age, hemoglobin, ESR, WBC, the differential count, RBC and platelets; HCT, red-cell indices, RDW-CV and platelet indices are absent. After removing eight duplicate records, 312 patients remained (67.6% positive). Platelet and WBC counts were converted to counts per microliter, one impossible RBC value (0) was set to missing, and missing values were imputed with Jamalpur medians. This cohort was never used for training, tuning or threshold selection.

Exploratory analysis. To avoid any influence of the test set, the descriptive analysis used the 1,208 training patients only (381 negative, 827 positive). Table II and Fig. 2 show that the two classes overlap heavily: positives had a slightly lower median platelet count (162 vs. 176 $\times 10^3$/µl) with a much wider spread, slightly lower WBC and HCT, and a lower RDW-CV, while several parameters differed only in their tails. No single CBC parameter separates the classes. Pairwise Spearman correlations among the 16 parameters were weak (largest $|\rho| = 0.35$, RBC with RDW-CV), and every correlation with dengue status was below 0.2 (largest: RDW-CV, $|\rho| = 0.19$; Fig. 3). Collinearity is therefore not a concern, but the weak univariate signal already suggests that discrimination from routine CBC values will be modest.

B. D-CHSS Feature Engineering

Each parameter x with clinical reference range $[L, U]$ (sex-specific for hemoglobin, RBC and hematocrit) is converted to

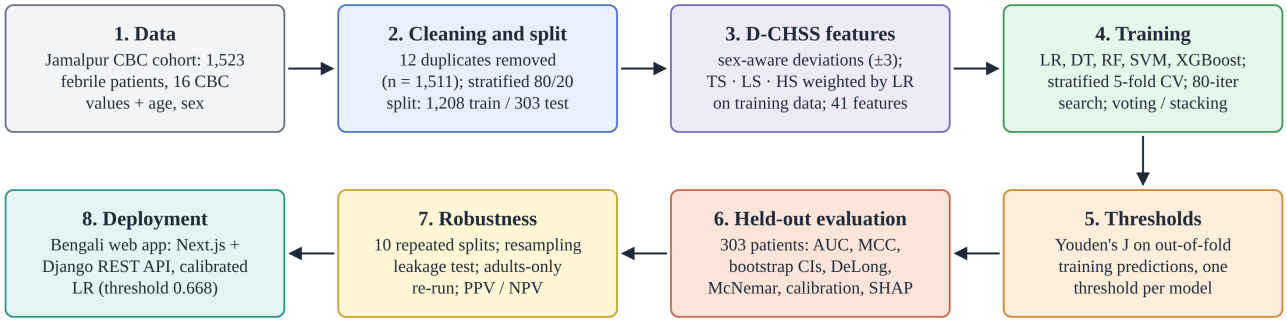


Fig. 1. End-to-end workflow of the study. Every learned step uses training data only; nothing is tuned on the held-out test set.

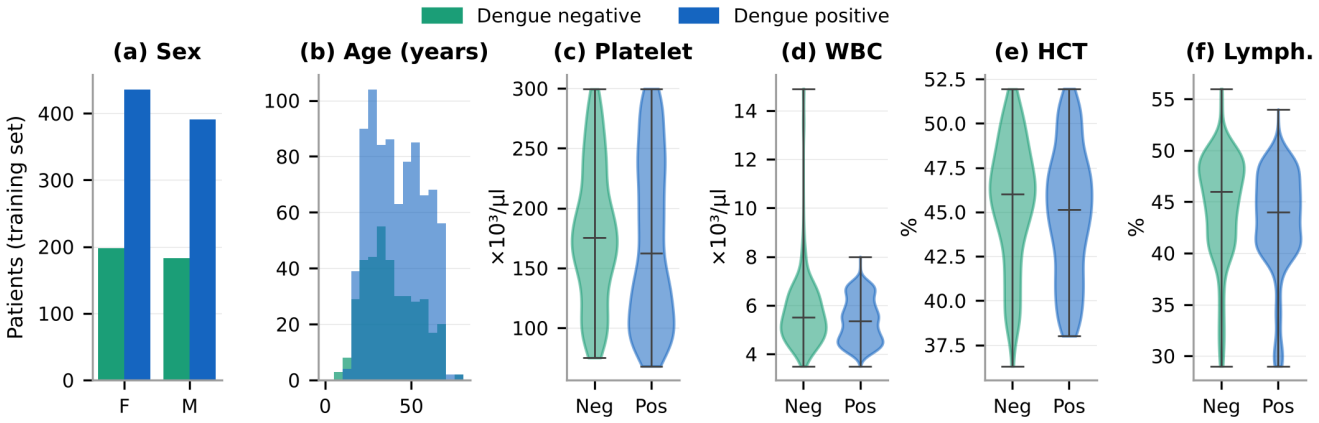


Fig. 2. Exploratory analysis of the training set: (a) sex and (b) age distribution by class; (c)–(f) violin plots of platelet count, WBC, HCT and lymphocytes. Horizontal lines mark medians and ranges. The classes overlap for every parameter.

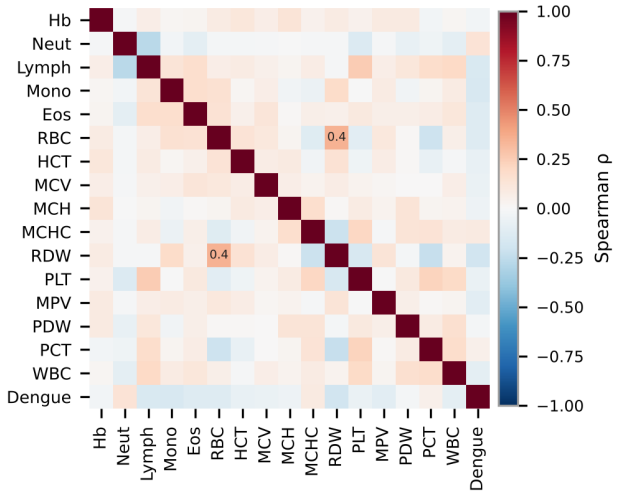


Fig. 3. Spearman correlation between the 16 CBC parameters and dengue status (training set). Values with $|\rho| \geq 0.3$ are printed.

a capped deviation

$$\mathrm{dev}(x) = \begin{cases} -(L-x)/L, & x < L \\ 0, & L \leq x \leq U \\ (x-U)/U, & x > U \end{cases}, \quad \mathrm{dev} \in [-3, 3]. \tag{1}$$

Because the ranges are fixed clinical constants, applying them to test data cannot leak information. Deviations are grouped by the clinical process they reflect: the Thrombocytopenia Score (TS: platelet, MPV, PDW, PCT), the Leukocyte Response Score (LS: WBC, neutrophils, lymphocytes, monocytes, eosinophils) and the Hemoconcentration Score (HS: HCT, hemoglobin, RBC). Each raw score is the mean absolute deviation of its parameters (Fig. 4). The scores are standardized on the training set, a logistic regression is fitted on them, and the absolute coefficients are normalized into weights:

$$w_k = \frac{|\beta_k|}{\sum_j |\beta_j|}, \qquad \text{D-CHSS} = \sum_{k \in \{TS, LS, HS\}} w_k \, \tilde{S}_k. \tag{2}$$

The learned coefficients were -0.128 (TS), -0.505 (LS) and -0.254 (HS), giving weights of 0.145, 0.569 and 0.287. Five ratios that need no extra test were added: NLR, PLR, LMR, PDW/MPV and RDW-CV/platelet. Four feature sets were compared: (A) raw CBC (16), (B) deviations (16), (C) D-CHSS components (4) and (D) the proposed full set (41).

C. Models, Threshold and Evaluation

Baselines were logistic regression (LR), decision tree (DT), Random Forest (RF) [22], SVM [23] and XGBoost [24] on raw CBC values, each tuned by grid search (Table III) with stratified five-fold cross-validation (CV). For the 41-feature

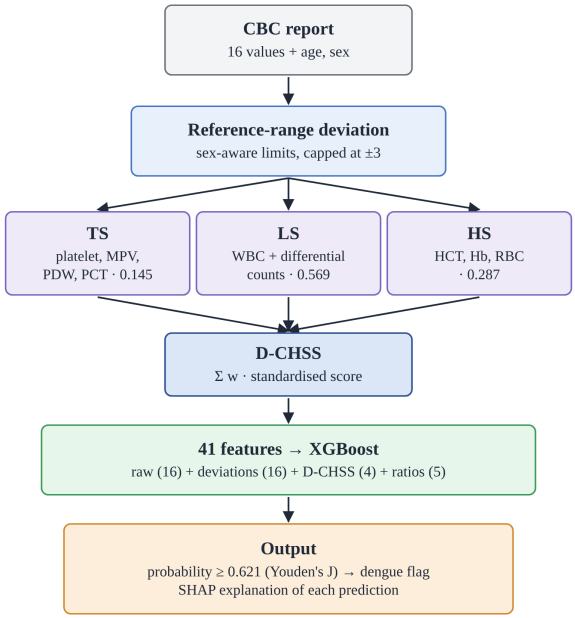


Fig. 4. Model overview. CBC values become capped, sex-aware deviations, which form the TS, LS and HS components and the weighted D-CHSS score; together with raw values and ratios they feed XGBoost, whose probability is compared with the training-derived threshold.

TABLE III
HYPERPARAMETER SEARCH SPACES AND SELECTED VALUES (TRAINING-SET CROSS-VALIDATION, SCORING = ROC-AUC)

Model	Search space	Selected
LR	$C \in \{0.01, 0.1, 1, 10\}$; class weight $\in$ {none, balanced}	$C = 0.01$, balanced
DT	depth $\in \{3, 5, 7, \text{none}\}$; min split $\in \{2, 5, 10\}$; min leaf $\in \{1, 2, 4\}$; class weight	depth 7, split 10, leaf 1, balanced
RF	trees $\in \{100, 200, 300\}$; depth $\in \{5, 10, \text{none}\}$; min split $\in \{2, 5\}$; max features $\in \{\sqrt{\cdot}, \log_2\}$; class weight	300 trees, depth 5, split 5, $\sqrt{\cdot}$, none
SVM (RBF)	$C \in \{0.1, 1, 10\}$; $\gamma \in$ {scale, auto}; class weight	$C = 0.1$, scale, none
XGBoost (raw)	trees $\in \{100, 200\}$; depth $\in \{3, 5\}$; $\eta \in \{0.01, 0.1\}$; subsample, colsample $\in \{0.8, 1\}$	100 trees, depth 3, $\eta = 0.01$, 0.8, 0.8
Ours (41 feat.)	80 random draws: trees 100–400; depth 3–8; η 0.01–0.15; subsample, colsample 0.6–1; min child weight 1–7; γ 0–0.3; scale_pos_weight $\in \{1, 0.46\}$	200 trees, depth 3, $\eta = 0.01$, subsample 0.6, colsample 0.8, min child 1, $\gamma = 0.1$, scale_pos_weight 1

set, XGBoost was tuned with an 80-iteration randomized search including `scale_pos_weight`; we preferred class weighting and threshold tuning to synthetic oversampling so that the test set contains only real patients. Soft-voting and stacking ensembles were also built. Algorithm 1 summarizes the protocol. The final model was chosen by training-CV ROC-AUC only, and the decision threshold was set by maximizing Youden's J [25] on out-of-fold training predictions (0.621) and then frozen. For a fair comparison, every baseline and ablation model received its own threshold by exactly the same procedure, so no model is judged at the default 0.5.

Algorithm 1 Leakage-safe D-CHSS pipeline

Require: cohort $\mathcal{D}$ of CBC records with confirmed labels
1: remove exact duplicates; split $\mathcal{D}$ into $\mathcal{D}_{tr}$ (80%) and $\mathcal{D}_{te}$ (20%), stratified
2: fit imputers and scalers on $\mathcal{D}_{tr}$ only
3: compute capped deviations (1) for $\mathcal{D}_{tr}$ and $\mathcal{D}_{te}$ with fixed reference ranges
4: on $\mathcal{D}_{tr}$: standardize TS, LS, HS; fit LR; set w_k by (2)
5: build the 41-feature set for both partitions
6: **for all** candidate models m **do**
7: tune m by stratified 5-fold CV on $\mathcal{D}_{tr}$
8: **end for**
9: select the model with the highest training-CV ROC-AUC
10: $t^* \leftarrow \arg\max_t J(t)$ on out-of-fold training predictions
11: refit on $\mathcal{D}_{tr}$; freeze model and t^*
12: evaluate once on $\mathcal{D}_{te}$; report metrics with bootstrap CIs

We report accuracy, sensitivity (recall), specificity, F1, ROC-AUC, PR-AUC, the Matthews correlation coefficient (MCC) [21] and the Brier score, together with calibration curves, decision curve analysis (DCA) [26], McNemar's exact test [27] and SHAP explanations [28], following TRIPOD+AI guidance [29]. A CV stability check refitted the scaler and D-CHSS weights inside every fold. All experiments used Python with scikit-learn [30], XGBoost and SHAP.

With TP, TN, FP and FN the cells of the confusion matrix,

$$\text{Sens} = \frac{TP}{TP+FN}, \quad \text{Spec} = \frac{TN}{TN+FP}, \quad F_1 = \frac{2TP}{2TP+FP+FN}, \tag{3}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}, \tag{4}$$

and the threshold maximizes Youden's $J(t) = \text{Sens}(t) + \text{Spec}(t) - 1$. Predictive values at prevalence π follow from Bayes' rule,

$$\text{PPV} = \frac{\text{Sens}\,\pi}{\text{Sens}\,\pi + (1-\text{Spec})(1-\pi)}, \quad \text{NPV} = \frac{\text{Spec}(1-\pi)}{\text{Spec}(1-\pi) + (1-\text{Sens})\pi}. \tag{5}$$

D. Uncertainty, Robustness and Leakage Checks

On the test set we report 95% confidence intervals (CIs) from 2,000 bootstrap resamples [31] and compare ROC-AUCs with DeLong's test [32]. Because a single split can be lucky or unlucky, we re-ran the entire pipeline (split, feature engineering, D-CHSS weights, hyperparameter search, model selection and Youden thresholds) for ten split seeds (42 and 1–9) and report mean $\pm$ SD. To measure how resampling affects reported accuracy, we compared four protocols on the same data with a stacking ensemble of LR, RF and XGBoost over five seeds: random oversampling of the whole dataset before the split, SMOTE [19] before the split, SMOTE inside the training set only, and no resampling. These are generic protocols, not a reproduction of any published pipeline. Because adult reference ranges do not fit children, we also re-ran the pipeline on the 1,473 adults only (38 patients, 2.5%, were younger than 18). Finally, positive and negative predictive values (PPV, NPV) were recomputed for prevalences of 20% and 40% from the test sensitivity and specificity.

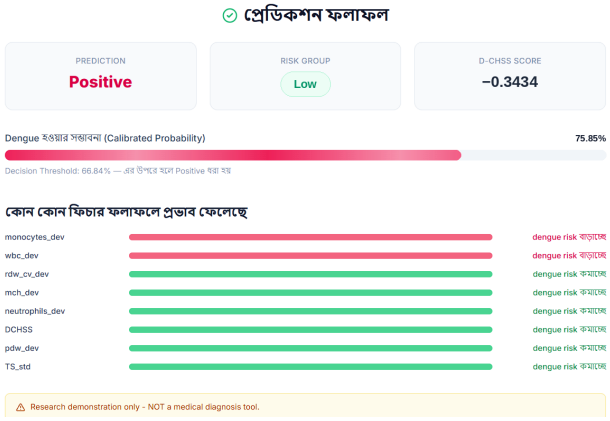


Fig. 5. Result page of the web application: class label, D-CHSS group, D-CHSS score, calibrated probability with the decision threshold, and features that raise (red) or lower (green) the probability.

E. External Validation

Only ten variables are shared by both cohorts (age, sex, hemoglobin, RBC, WBC, neutrophils, lymphocytes, monocytes, eosinophils and platelets), so the 41-feature model cannot be applied directly. We therefore rebuilt the pipeline on these variables using all 1,511 Jamalpur patients. The same reference ranges and ± 3 cap gave eight deviations; D-CHSS components were formed from the available parameters (TS: platelet; LS: WBC and the four differential counts; HS: hemoglobin and RBC) with weights learned in Jamalpur (0.216, 0.441, 0.342); and NLR, PLR and LMR were added, giving 25 features. LR, RF and XGBoost on the eight raw CBC values, XGBoost on the four D-CHSS components and XGBoost on the 25-feature set were tuned by five-fold CV in Jamalpur, and each threshold was fixed at the Youden point of its out-of-fold Jamalpur predictions. The models were then applied once, unchanged, to the Dhaka cohort. As references that need no training, we report the platelet count alone and the rule platelet $< 150 \times 10^3/\mu$L, and, to show what a local model can reach, a Random Forest cross-validated within Dhaka.

F. Web Application

The framework is deployed as the *D-CHSS Dengue Predictor* [33]. For a small serverless footprint, the deployed model is a logistic regression with isotonic calibration [34] implemented in pure Python, using the same reference ranges and D-CHSS transformations. It returns a calibrated probability, the D-CHSS group (labelled "risk group" in the app) and feature-level contributions (Fig. 5). Testing the app model on the test set revealed two feature calculations that did not match the training pipeline; both were corrected before evaluation.

IV. Results

A. Baselines and Ablation

Tuning raised the training-CV ROC-AUC of every baseline. On the test set, with each model at its own Youden threshold (Table IV), all models found most dengue cases but rejected fewer than half of the non-dengue patients. RF had the highest baseline ROC-AUC (0.663). RF and raw-CBC XGBoost reached higher specificity than our model (45.3% and 43.2% vs. 36.8%) but missed more dengue cases (recall 88.9% and 90.9% vs. 95.2%); our model kept the highest ROC-AUC, PR-AUC, F1 and MCC. Its advantage is therefore in ranking and overall balance, not in specificity. With identical hyperparameters and per-model thresholds (Table V), deviations alone (B) and the four D-CHSS features alone (C) were worse than raw CBC (A), while combining everything (D) gave the best ROC-AUC and PR-AUC with the same MCC as raw CBC. The engineered features therefore improve how patients are ranked rather than the decision at one threshold, and they complement the raw measurements rather than replace them.

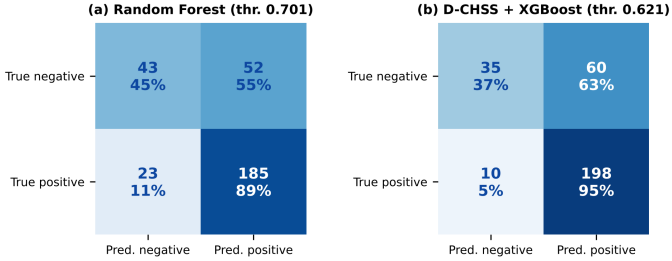


Fig. 6. Confusion matrices on the test set ($n = 303$), each model at its own Youden threshold.

TABLE IV
TEST-SET PERFORMANCE (303 PATIENTS). EVERY MODEL USES ITS OWN YOUDEN THRESHOLD FROM OUT-OF-FOLD TRAINING PREDICTIONS (LR 0.444, DT 0.488, RF 0.701, SVM 0.637, XGBOOST 0.686, OURS 0.621); THE WEB APP USES 0.668.

Model	Recall	Spec.	F1	AUC	MCC
Logistic regression	0.885	0.337	0.809	0.622	0.265
Decision tree	0.923	0.368	0.835	0.633	0.361
Random Forest	0.889	0.453	0.831	0.663	0.384
SVM (RBF)	0.938	0.347	0.839	0.658	0.368
XGBoost (raw CBC)	0.909	0.432	0.838	0.658	0.396
D-CHSS + XGBoost	0.952	0.368	**0.850**	**0.685**	**0.418**
Web-app LR (calibrated)	0.875	0.484	0.829	0.711	0.392

TABLE V
ABLATION ON THE TEST SET (XGBOOST, IDENTICAL HYPERPARAMETERS, EACH AT ITS OWN YOUDEN THRESHOLD)

Feature set	Recall	Spec.	AUC	PR-AUC	MCC
A: Raw CBC (16)	0.957	0.358	0.653	0.762	**0.418**
B: Deviation (16)	0.952	0.295	0.633	0.753	0.346
C: D-CHSS (4)	0.784	0.463	0.604	0.722	0.251
D: Full (41)	0.952	0.368	**0.685**	**0.800**	**0.418**

Thresholds: A 0.626, B 0.621, C 0.685, D 0.621.

B. Final Model

The single XGBoost model reached a training-CV ROC-AUC of 0.696 $\pm$ 0.055, ahead of the stacking (0.692) and voting (0.687) ensembles, and was selected. Compared with the default 0.5, the Youden threshold raised specificity from

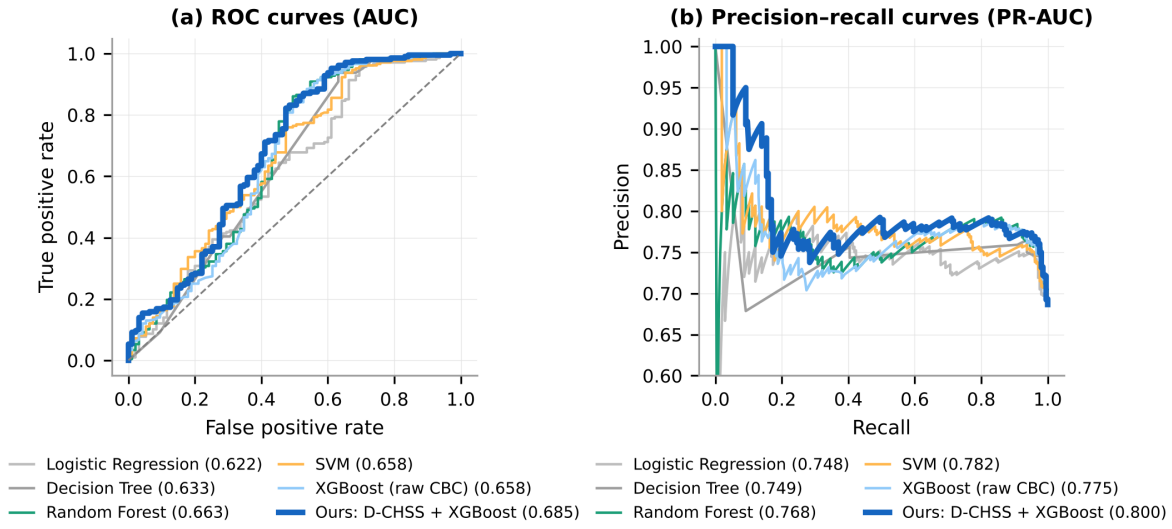


Fig. 7. ROC and precision–recall curves of the baselines and the proposed model on the test set (legend values: AUC and PR-AUC).

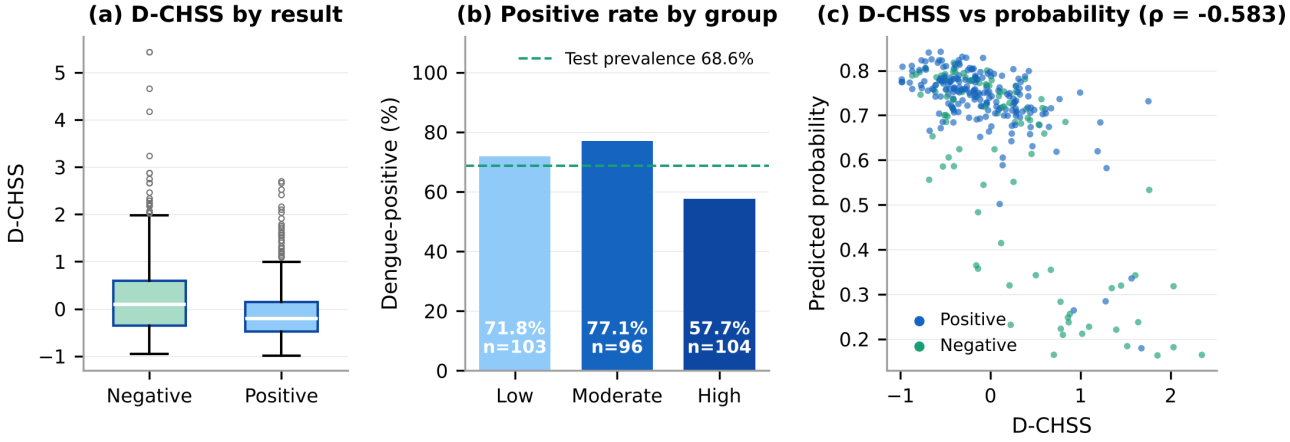


Fig. 8. Behaviour of the D-CHSS signature: (a) all 1,511 patients; (b), (c) the 303 test patients.

28.4% to 36.8% while recall fell only from 98.1% to 95.2%. On the test set the model reached 76.9% accuracy, 76.7% precision, 95.2% recall, 36.8% specificity, F1 0.850, ROC-AUC 0.685, PR-AUC 0.800, MCC 0.418 and Brier score 0.178. The leakage-safe CV estimates were close (ROC-AUC 0.696, F1 0.853), suggesting the model is not overfitted. It identified 198 of 208 dengue cases and correctly ruled out 35 of 95 negatives (Fig. 6); compared with RF at its own threshold it missed 13 fewer dengue cases (10 vs. 23) at the cost of eight more false alarms (60 vs. 52), and it had the largest areas under both the ROC and precision–recall curves (Fig. 7). Fig. 10 compares the four strongest models on six metrics. The deployed web-app model caught 182 of 208 cases (87.5%) and ruled out 46 of 95 negatives (48.4%), with a ROC-AUC of 0.711.

C. Uncertainty and Repeated Splits

On the seed-42 test set the final model had a sensitivity of 95.2% (95% CI 92.2–97.7%), a specificity of 36.8% (27.4–46.7%) and an MCC of 0.418 (0.307–0.525). The ROC-AUC intervals of all models overlap widely (Table VI); only LR was significantly worse than ours (DeLong $p = 0.022$). Over ten repeated splits (Fig. 11), our pipeline had the highest mean ROC-AUC, PR-AUC (0.791 ± 0.024), MCC (0.437 ± 0.055) and sensitivity ($94.7\% \pm 1.6$), but it beat RF in only 7 and raw-CBC XGBoost in 5 of the ten splits, by a mean of 0.003 and 0.005, far less than the split-to-split SD, and it was the best model in 4 splits. The pipeline chose the full-feature XGBoost in eight splits and the stacking ensemble in two. The seed-42 result is therefore typical rather than lucky, but the advantage over strong baselines is within noise. With identical hyperparameters, the four D-CHSS features alone averaged a ROC-AUC of 0.674 ± 0.035 and a PR-AUC of 0.790, slightly above the sixteen raw values (0.668 ± 0.039, 0.773); seed 42 was the least favorable split for the signature (0.604).

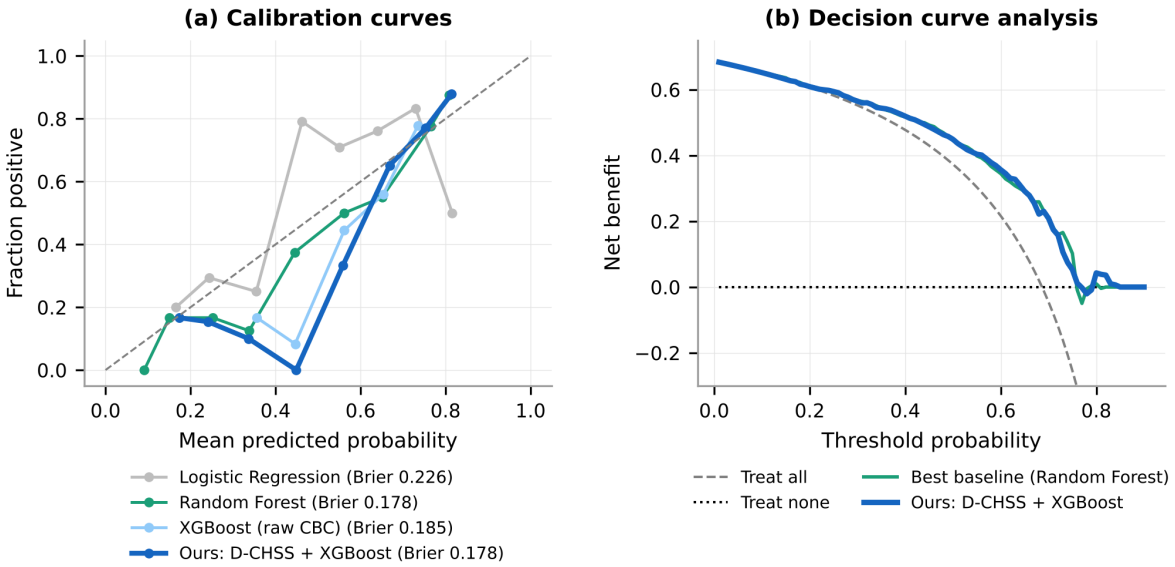


Fig. 9. Calibration curves with Brier scores and decision curve analysis on the test set.

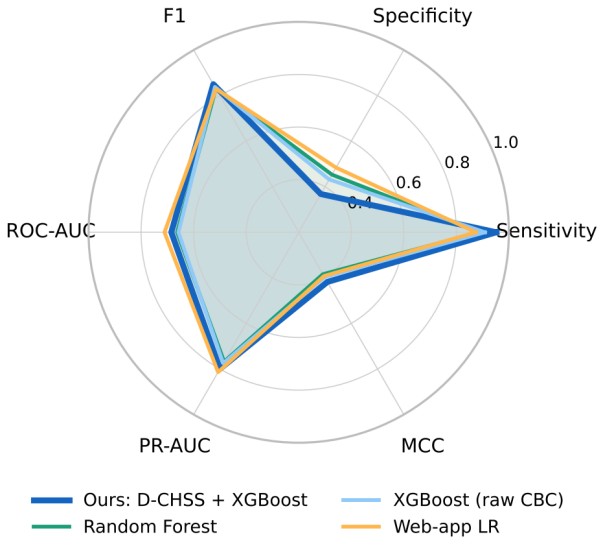


Fig. 10. Test-set profile of the strongest models, each at its own threshold. Our model leads on sensitivity, F1, PR-AUC and MCC; RF, raw-CBC XGBoost and the web-app LR trade sensitivity for specificity.

TABLE VI
UNCERTAINTY: SEED-42 TEST ROC-AUC WITH 95% BOOTSTRAP CI, DELONG TEST AGAINST OUR MODEL, AND MEAN ± SD OVER TEN REPEATED SPLITS

Model	Test AUC (95% CI)	DeLong p	10-split AUC	10-split MCC
LR	0.622 (0.550–0.695)	0.022	0.642 ± 0.040	0.335 ± 0.064
DT	0.633 (0.563–0.703)	0.169	0.636 ± 0.044	0.353 ± 0.052
RF	0.663 (0.589–0.733)	0.229	0.685 ± 0.032	0.407 ± 0.039
SVM	0.658 (0.587–0.726)	0.332	0.678 ± 0.028	0.362 ± 0.038
XGBoost (raw)	0.658 (0.583–0.735)	0.174	0.683 ± 0.030	0.413 ± 0.044
Ours	**0.685** (0.615–0.757)	–	**0.688 ± 0.030**	**0.437 ± 0.055**
Web-app LR	0.711 (0.646–0.777)	0.226	–	–

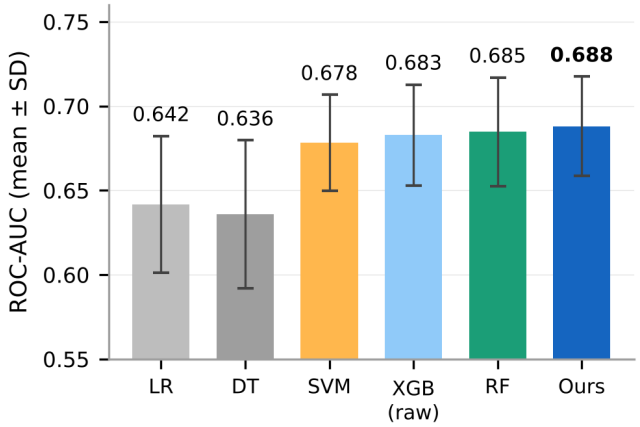


Fig. 11. Test ROC-AUC over ten repeated splits that re-ran the whole pipeline (mean ± SD). All strong models reach about 0.69; the differences are smaller than the split-to-split variation.

D. Effect of Resampling Before the Split

Resampling before the split places copies or interpolations of the same patients in both sets. On the deduplicated data (Fig. 12), random oversampling before the split raised accuracy to 92.4% and ROC-AUC to 0.956, and SMOTE before the split to 83.5% and 0.882, whereas the two leak-free protocols stayed near 75% and 0.67, in line with our results and with [10]. Results on the original 1,523 records differed by at most about 0.01. These experiments do not show that any published study, including [9], was affected by leakage: its exact pipeline is not available to us, and other choices could explain different results. They show only that, on this cohort and with our models, accuracies above about 80% appeared only when resampling preceded the split. The external cohort also shows that high accuracy is not by itself a sign of leakage (Section IV-E).

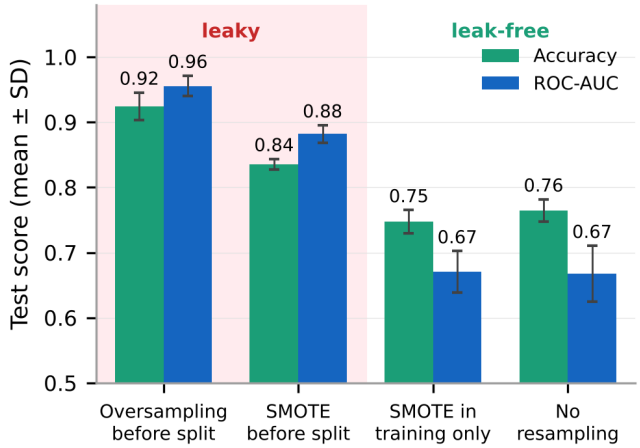


Fig. 12. Effect of the resampling protocol on a stacking ensemble (deduplicated data, five seeds, mean ± SD, threshold 0.5). Resampling before the split inflates both accuracy and ROC-AUC.

TABLE VII
EXTERNAL VALIDATION ON THE DHAKA COHORT ($n = 312$). MODELS WERE DEVELOPED ON ALL JAMALPUR PATIENTS WITH THE TEN SHARED VARIABLES; SENSITIVITY AND SPECIFICITY USE THE JAMALPUR YOUDEN THRESHOLD.

Model	Int. CV	Ext. AUC [95% CI]	Sens.	Spec.	MCC
LR (raw, 8)	0.640	0.314 [0.252, 0.375]	45.5	19.8	−0.328
RF (raw, 8)	0.695	0.691 [0.628, 0.756]	39.3	66.3	0.055
XGBoost (raw, 8)	0.685	0.593 [0.515, 0.672]	13.7	89.1	0.040
D-CHSS comp. (4)	0.682	0.849 [0.800, 0.891]	89.1	63.4	0.547
D-CHSS set (25)	**0.704**	**0.888** [0.846, 0.923]	67.8	91.1	**0.551**
Platelet alone[a]	0.528	0.954 [0.931, 0.973]	93.4	72.3	0.685
Local RF (Dhaka CV)	–	0.926	–	–	–

[a] AUC of the platelet count; Sens./Spec./MCC for platelet $< 150 \times 10^3/\mu$L.

E. External Validation

The two cohorts differed sharply (Fig. 13a). In Jamalpur, the median platelet count was $163 \times 10^3/\mu$L in positives and $172 \times 10^3/\mu$L in negatives, and all values lay between 56 and $300 \times 10^3/\mu$L; in Dhaka it was $35 \times 10^3/\mu$L in positives and $201 \times 10^3/\mu$L in negatives. Dhaka positives also had lower WBC (median 5,680 vs. 7,610/μL) and neutrophils (53% vs. 67%) and more lymphocytes (36% vs. 25%), the textbook dengue picture, whereas these medians were almost identical between the Jamalpur classes. Internally, the reduced models reached CV ROC-AUCs of 0.64–0.70, close to the full models (Table VII).

Externally, models trained on raw values transferred poorly: RF reached 0.691, XGBoost 0.593 and LR 0.314, i.e., LR ranked Dhaka patients in the wrong order. The four D-CHSS components alone reached 0.849 (95% CI 0.800–0.891) and the 25-feature D-CHSS set 0.888 (0.846–0.923), both significantly above RF and XGBoost on raw values (DeLong $p < 10^{-7}$). A plausible reason is that tree models cannot extrapolate beyond their training range: two-thirds of Dhaka positives (67.3%) had platelet counts below the Jamalpur minimum, which raw-value trees treat like the lowest training values, while LR learned an almost flat platelet slope in Jamalpur and neutrophil and monocyte slopes opposite to the pattern seen in Dhaka. A deviation from the fixed $150 \times 10^3/\mu$L limit, by contrast, grows with the drop and keeps the same clinical meaning in both hospitals. Thresholds transferred less well than rankings: at its Jamalpur threshold the 25-feature model had 67.8% sensitivity and 91.1% specificity, the reverse of its internal trade-off, so a new site would need to recalibrate the threshold.

The simplest reference was the strongest. The platelet count alone reached 0.954 (0.931–0.973) in Dhaka, significantly above the best transferred model (DeLong $p < 10^{-4}$), but only 0.528 in Jamalpur; the rule platelet $< 150 \times 10^3/\mu$L gave 93.4% sensitivity and 72.3% specificity in Dhaka, and a Random Forest cross-validated within Dhaka reached 0.926 (Fig. 13b). The D-CHSS score itself was not inversely related to dengue in Dhaka (ROC-AUC 0.564).

F. Behaviour of the D-CHSS Signature

D-CHSS differed significantly between the classes (Mann–Whitney $U = 179{,}822.5$, $p < 10^{-6}$, $r \approx 0.27$), but in the opposite direction to our hypothesis: negative patients had higher scores (Fig. 8a), and its Spearman correlation with the predicted probability was -0.583 (Fig. 8c). In tertile groups learned on the training set, the positive rate was 71.8% (Low), 77.1% (Moderate) and only 57.7% (High) against a prevalence of 68.6% (Fig. 8b). The inverse direction appeared in all ten repeated splits and in the adults-only re-run, and LS always received the largest weight (0.48–0.55), so it is a stable property of this cohort rather than an artifact of one split; in the Dhaka cohort the direction was not inverted (Section IV-E), which suggests a property of the Jamalpur case mix rather than of the signature itself. In its current form, D-CHSS measures how abnormal a blood picture is, not how likely dengue is, so its groups must not be read as dengue-risk categories.

G. Calibration and Clinical Usefulness

The tree models were reasonably calibrated: RF (Brier 0.178) and the proposed model (0.178) scored best, whereas LR (0.226) was poorly calibrated (Fig. 9a). In decision curve analysis (Fig. 9b), both the proposed model and RF give more net benefit than "treat all" for threshold probabilities between roughly 0.3 and 0.75, and their curves almost coincide, so their clinical usefulness is similar.

H. Explainability with SHAP

The most influential features were D-CHSS, platelet count and RDW-CV, followed by LS_std, NLR, LMR, monocytes, PLR and HS_std (Fig. 15); four of the top ten are signature features and three are ratios. High D-CHSS, RDW-CV, LS_std and monocyte values pushed predictions toward negative, whereas low platelets and low PLR pushed toward positive, matching the role of thrombocytopenia in dengue [1], [16]. For individual patients, SHAP shows the same picture: test patient 0, actually negative, received a probability of 0.817

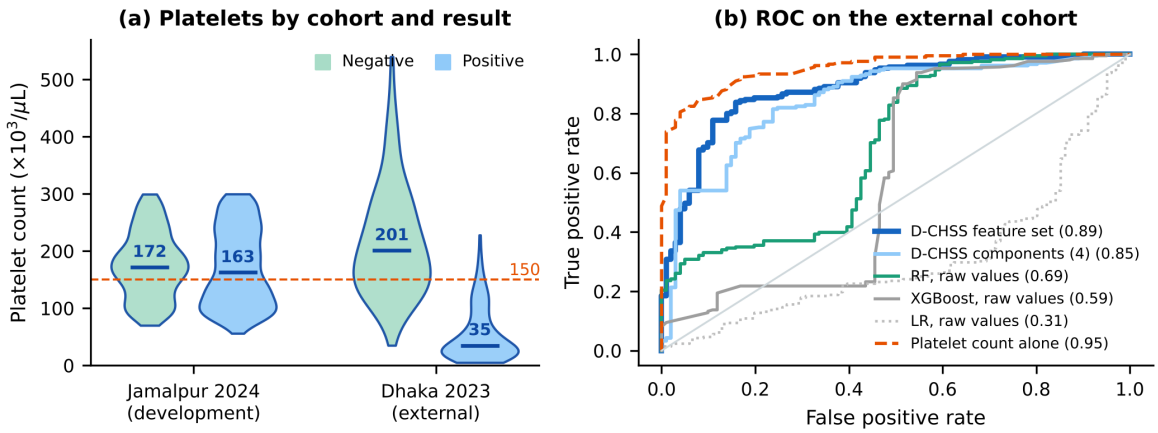


Fig. 13. External validation. (a) Platelet counts by cohort and result (median labels, dashed line: lower reference limit). (b) ROC curves on the Dhaka cohort for models developed in Jamalpur and for the platelet count alone (legend: ROC-AUC).

mainly because of a low platelet count (109,328/cumm), a false positive caused by thrombocytopenia without dengue. Fig. 14 shows two correct predictions on the log-odds scale. For a true positive ($p = 0.842$), a platelet count of 72,000/µl and high PLR and NLR moved the score toward dengue; for a true negative ($p = 0.163$), a high LS score, a raised RDW-CV (15.8%) and a high D-CHSS moved it away. Such per-patient explanations are what the web application shows to the pathologist.

I. Significance, Prior Work and Sensitivity Analysis

McNemar's exact test against RF, both at their own thresholds, on the same 303 patients was not significant (both correct 217, both wrong 59, ours alone correct 16, RF alone 11; $p = 0.442$), so the better metrics are encouraging but not conclusive. Our ROC-AUC of 0.685 is almost identical to the 0.69 of Hasan *et al.* [10] on the same cohort, which suggests that the information in routine CBC data from this cohort is genuinely limited; we could reach the much higher accuracy reported in [9] only when resampling preceded the split (Fig. 12), which is consistent with, but does not establish, a difference in protocol.

Because the main signature uses absolute deviations and absolute weights, we tested a direction-aware version in which each parameter received a dengue direction fixed in advance from the literature [1], [7], [16] and the weights kept their sign. All other steps, including re-tuning and the Youden threshold (0.425), followed the main pipeline. The score then pointed the right way (median 0.17 in positive vs. -0.12 in negative patients; Spearman $+0.284$), but discrimination did not improve (Table VIII).

TABLE VIII
MAIN (ABSOLUTE) VS. DIRECTION-AWARE (SIGNED) D-CHSS ON THE TEST SET

Metric	Absolute (main)	Signed
ROC-AUC / PR-AUC	0.685 / 0.800	0.671 / 0.786
Sensitivity / specificity	95.2% / 36.8%	94.2% / 38.9%
MCC / Brier	0.418 / 0.178	0.418 / 0.205
5-fold CV ROC-AUC	0.696	0.694
Spearman (D-CHSS vs. prob.)	-0.583	$+0.284$
Positive rate Low/Mod./High	71.8/77.1/57.7%	63.9/77.1/64.4%

V. DISCUSSION

What the results show. In the Jamalpur cohort, routine CBC values detect dengue only moderately, and this ceiling is stable: about 0.69 ROC-AUC across ten splits and all strong models. With about 95% sensitivity and 37% specificity (likelihood ratios $\mathrm{LR}^+ = 1.51$, $\mathrm{LR}^- = 0.13$), a negative result lowers the probability of dengue considerably, whereas a positive result raises it only a little. The cohort is 68.6% positive, far more than febrile patients in general. If sensitivity and specificity carry over, at 20% prevalence the PPV would be 27.4% (95% CI 24.7–30.9%) and the NPV 96.8% (94.5–98.6%); at 40% they would be 50.1% and 92.0%. Most positive flags would then be false alarms, so the tool can help identify patients unlikely to have dengue but cannot confirm dengue, and every positive result needs NS1 or serology; spectrum differences between this hospital cohort and primary care may also lower sensitivity. D-CHSS helps as a compact complement: its four interpretable features carry about the same ranking information as the sixteen raw values, and within the full set it became the most important SHAP feature, although the full model's gain over strong baselines is within noise. The deployed logistic regression had a slightly higher test ROC-AUC (0.711) than XGBoost (0.685), but the difference was not significant (DeLong $p = 0.226$). XGBoost remains the main model because it was chosen by training-CV ROC-AUC before the test set was examined; the web-app model was built later for deployment and took no part in that choice. Promoting whichever model scores best on the test set would turn the test set into a selection set and bias the estimate upward.

What the external cohort adds. The ceiling of about 0.69

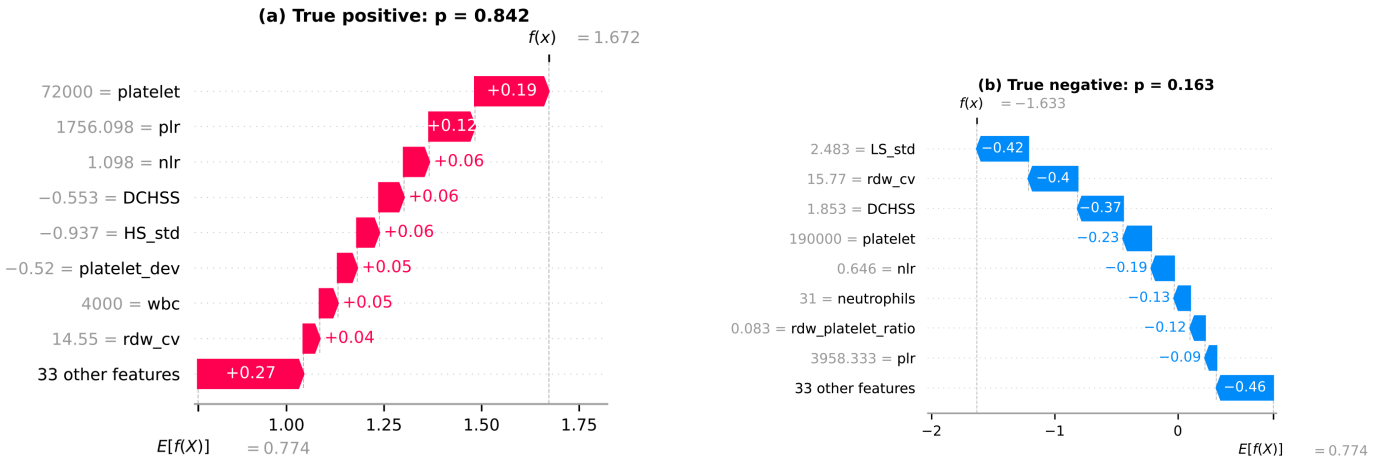


Fig. 14. Local SHAP explanations (log-odds) for (a) a true positive and (b) a true negative test patient. Red bars push toward dengue, blue bars away from it; $E[f(X)]$ is the average model output.

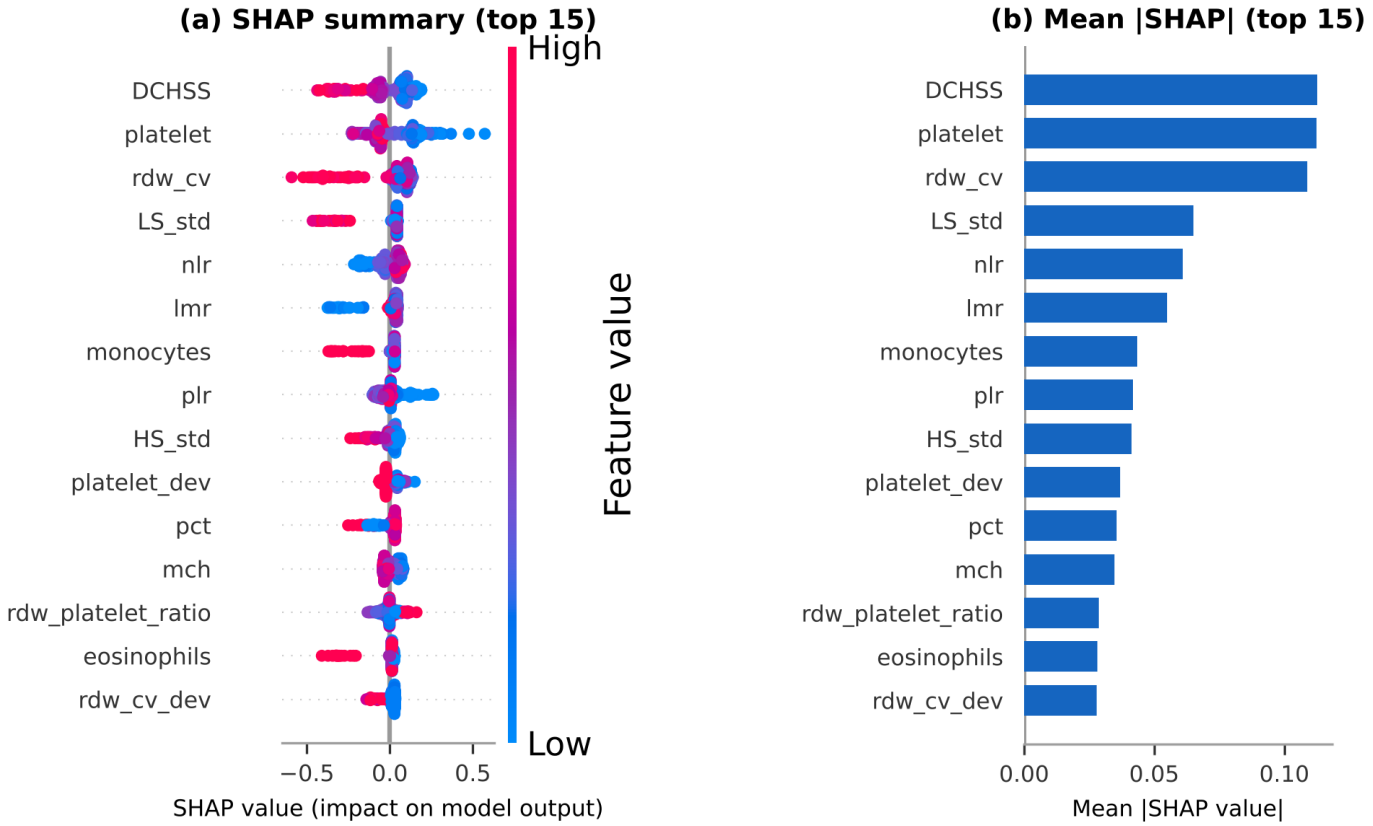


Fig. 15. Global SHAP explanations of the final model (top 15 of 41 features): (a) direction and size of each feature's effect per patient; (b) mean absolute SHAP value.

is a property of the Jamalpur case mix, not of the CBC itself. Where dengue patients present with marked thrombocytopenia, as in the Dhaka hospitals during the 2023 outbreak, a single parameter separates the classes well and machine learning adds little over the platelet rule; where they do not, many models together cannot exceed about 0.7. The high accuracy of Riya *et al.* [8] is consistent with the strong platelet signal in their cohort. Jamalpur positives were barely thrombocytopenic and all Jamalpur platelet counts lay within 56–300 $\times$ $10^3/\mu$L, which is unusual for hospital dengue; it may reflect early, NS1-detected illness or the way records were selected, which we cannot verify. Three practical lessons follow: reported performance should be read together with the case mix (at least the platelet distribution and, ideally, illness day); models should be validated in another hospital before deployment; and features anchored to fixed clinical reference

ranges, such as the D-CHSS deviations, transfer between sites better than raw values, which is the main practical value of D-CHSS.

Why is D-CHSS higher in negative patients? The sensitivity analysis shows that discarding the sign was not the main cause. On the training set, only the platelet count clearly moved in its textbook direction (47.8% of positives below range vs. 30.2% of negatives). WBC was low in only 0.3% of patients, so the classic leukopenia of dengue cannot be seen through reference-range deviations here, while relative lymphocytosis (82.2% vs. 76.8%), low neutrophils (16.0% vs. 1.8%), raised HCT (43.6% vs. 37.2%) and raised RBC (24.1% vs. 7.6%) were more common in negatives. The negatives were febrile patients with other infections that also disturb the blood count [7], [15], so "abnormal" does not mean "dengue" in this cohort; label noise from NS1 weakening after the first days may add to this [2].

Limitations, bias and ethics. The test set was not used only once: after the main evaluation it was used again to evaluate the web-app model, which revealed two feature-calculation errors that were fixed before that model was re-evaluated, and for the sensitivity analysis and the equal-threshold comparison. No model or threshold was tuned on test results, but these repeated looks mean the test figures, especially for the web-app model, are slightly less independent than a single untouched evaluation. All data come from one hospital and one season with 68% positives. The public dataset contains no test dates, so a temporal split (e.g., training on February–June and testing on July–September) was impossible. The external cohort shares only ten variables, so neither the 41-feature model nor the web-app model could be tested on it; it is small (312 patients), comes from another study whose inclusion criteria and confirmation tests differ from ours, and illness day is unknown in both cohorts, so differences in case mix cannot be separated from differences in selection. Illness day, symptoms and serial CBCs were unavailable [16]. Adult reference ranges were applied to the 38 patients younger than 18; re-running the whole pipeline on adults only (1,178 training and 295 test patients) gave a ROC-AUC of 0.664, within the split-to-split range, without changing the inverse direction of D-CHSS, but pediatric ranges should be used in future work. The deployed model is a logistic regression rather than the research XGBoost. Selection, label and measurement biases may shift probabilities at other sites, and a precise percentage may be over-trusted despite low specificity [12]. The tool is therefore a screening aid only: every result should be confirmed by NS1 or serology, the decision stays with the clinician, and new data require ethics approval, informed consent and coded identifiers.

Contribution to the SDGs. The system mainly supports SDG 3 [35]: dengue is a WHO neglected tropical disease [36], and flagging likely cases from the first routine blood test can shorten the time to clinical attention (target 3.3) while aggregated screens could serve as an early outbreak signal (3.d). Because it needs only a CBC analyzer, a browser and a Bengali interface, it can bring the same decision support to district and upazila hospitals (SDG 10, targets 10.2–10.3) [37], and it builds local AI capacity on Bangladeshi data (SDG 9) [38]. Because performance depended strongly on case mix in our external validation, the model and its threshold must be validated locally before use at any new site.

VI. Conclusion and Future Work

Routine CBC data carry a real but cohort-dependent dengue signal. On 1,511 Jamalpur patients, every strong model reached the same leakage-free ceiling of about 0.69 ROC-AUC over ten repeated splits, our pipeline's advantage over Random Forest and raw-CBC XGBoost was within noise, and accuracies above about 80% appeared only when resampling preceded the train/test split. On an independent Dhaka cohort the signal was much stronger: the platelet count alone reached 0.95, models built on raw Jamalpur values transferred poorly, and features anchored to fixed reference ranges transferred well (0.89). D-CHSS is therefore most useful as an interpretable and transportable component rather than as a stand-alone risk score: its four features matched the sixteen raw values internally, but its direction differed between cohorts. A negative result from the Jamalpur model is fairly reassuring, but a positive result must be confirmed. Future work will (i) collect multi-center data with illness day and the full CBC panel, so that the complete model can be validated externally and temporally; (ii) rebuild D-CHSS from parameter directions confirmed in the target population; (iii) add symptoms and serial CBCs with age-specific ranges; and (iv) extend the application with conformal uncertainty estimates [39] and input-range checks.

Data and Code Availability

The Dengue Fever Hematological Dataset is publicly available from Mendeley Data [20], and the external Dhaka cohort from the repository of Riya *et al.* [8] (https://github.com/mritunjoychk17/Dengue-Prediction-in-Bangladesh-Using-Machine-Learning). The notebook and analysis code are available at https://github.com/shifat35/dchss-dengue-detection, and the web application at https://dengue-two-delta.vercel.app/.

Acknowledgment

The authors thank Rashik Rahman, Department of Computer Science and Engineering, University of Asia Pacific, for his guidance in the CSE 314 course, and the creators of both public datasets for making them available.

Declaration of AI Usage

An AI assistant (Claude, Anthropic) was used to help shorten and summarize our draft, refine its language, search for related papers and check reference details; to write and run the code for the direction-aware sensitivity analysis, the equal-threshold comparison, the bootstrap and DeLong analyses, the ten repeated splits, the resampling-protocol experiment, the adults-only re-run, the external validation on the Dhaka cohort and the predictive values; to re-run our notebook unchanged to

redraw the figures at print size; and to adapt the report into this IEEE format. The original analysis code and web application were written by the group, and AI was used there only for debugging. All authors reviewed the text and results and take full responsibility for the content.

REFERENCES

[1] World Health Organization, *Dengue: Guidelines for Diagnosis, Treatment, Prevention and Control*, new ed. Geneva: WHO, 2009. https://www.who.int/publications/i/item/9789241547871

[2] M. G. Guzman and E. Harris, "Dengue," *The Lancet*, vol. 385, no. 9966, pp. 453–465, 2015. https://doi.org/10.1016/S0140-6736(14)60572-9

[3] S. Bhatt *et al.*, "The global distribution and burden of dengue," *Nature*, vol. 496, pp. 504–507, 2013. https://doi.org/10.1038/nature12060

[4] N. Sharif *et al.*, "Evolving epidemiology, clinical features, and genotyping of dengue outbreaks in Bangladesh, 2000–2024: a systematic review," *Front. Microbiol.*, vol. 15, Art. 1481418, 2024. https://doi.org/10.3389/fmicb.2024.1481418

[5] M. A. B. Shiddik, F. Z. Toshi, S. Yesmin, and M. S. Rahman, "District-level dengue early warning prediction system in Bangladesh using hybrid explainable AI and Bayesian deep learning," *Trop. Med. Infect. Dis.*, vol. 11, no. 3, Art. 73, 2026. https://doi.org/10.3390/tropicalmed11030073

[6] M. E. Haque, M. Nurul Absur, F. Al Farid, J. Uddin, and H. Abdul Karim, "A novel interpretable and real-time dengue prediction framework using clinical blood parameters with genetic and GAN-based optimization," *Front. Artif. Intell.*, vol. 8, Art. 1626699, 2025. https://doi.org/10.3389/frai.2025.1626699

[7] J. A. Potts and A. L. Rothman, "Clinical and laboratory features that distinguish dengue from other febrile illnesses in endemic populations," *Trop. Med. Int. Health*, vol. 13, no. 11, pp. 1328–1340, 2008. https://doi.org/10.1111/j.1365-3156.2008.02151.x

[8] N. J. Riya, M. Chakraborty, and R. Khan, "Artificial intelligence-based early detection of dengue using CBC data," *IEEE Access*, vol. 12, pp. 112355–112367, 2024. https://doi.org/10.1109/ACCESS.2024.3443299

[9] I. A. Tuhin, A. K. M. F. K. Siam, M. M. R. Shanto, M. R. Mia, I. Mahmud, and A. Ghosh, "An interpretable machine learning model for dengue detection with clinical hematological data," *Healthcare Analytics*, vol. 8, Art. 100430, 2025. https://doi.org/10.1016/j.health.2025.100430

[10] M. M. I. Hasan, A. Habib, S. B. Sikto, and M. M. Ahmmed, "A novel hematological machine learning framework for predictive modeling of dengue diagnosis using CBC parameters in Bangladesh," in *Proc. IDAA 2025*, Adv. Intell. Syst. Res., vol. 206, 2026. https://doi.org/10.2991/978-94-6239-664-7_12

[11] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, Art. 100804, 2023. https://doi.org/10.1016/j.patter.2023.100804

[12] J. Amann, A. Blasimme, E. Vayena, D. Frey, and V. I. Madai, "Explainability for artificial intelligence in healthcare: a multidisciplinary perspective," *BMC Med. Inform. Decis. Mak.*, vol. 20, Art. 310, 2020. https://doi.org/10.1186/s12911-020-01332-6

[13] S. K. Dey *et al.*, "Prediction of dengue incidents using hospitalized patients, metrological and socio-economic data in Bangladesh: A machine learning approach," *PLOS ONE*, vol. 17, no. 7, e0270933, 2022. https://doi.org/10.1371/journal.pone.0270933

[14] B. Liu, M. F. Hossain, and S. Hossain, "A comparative evaluation of multiple machine learning approaches for forecasting dengue outbreaks in Bangladesh," *Sci. Rep.*, vol. 15, Art. 35931, 2025. https://doi.org/10.1038/s41598-025-19752-7

[15] R. P. Daumas *et al.*, "Clinical and laboratory features that discriminate dengue from other febrile illnesses: a diagnostic accuracy study in Rio de Janeiro, Brazil," *BMC Infect. Dis.*, vol. 13, Art. 77, 2013. https://doi.org/10.1186/1471-2334-13-77

[16] K. D. Rosenberger *et al.*, "Early diagnostic indicators of dengue versus other febrile illnesses in Asia and Latin America (IDAMS study): a multicentre, prospective, observational study," *Lancet Glob. Health*, vol. 11, no. 3, pp. e361–e372, 2023. https://doi.org/10.1016/S2214-109X(22)00514-9

[17] T.-S. Ho *et al.*, "Comparing machine learning with case-control models to identify confirmed dengue cases," *PLoS Negl. Trop. Dis.*, vol. 14, no. 11, e0008843, 2020. https://doi.org/10.1371/journal.pntd.0008843

[18] A. Qaiser *et al.*, "Support vector machine outperforms other machine learning models in early diagnosis of dengue using routine clinical data," *Adv. Virol.*, vol. 2024, Art. 5588127, 2024. https://doi.org/10.1155/2024/5588127

[19] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002. https://doi.org/10.1613/jair.953

[20] "Dengue Fever Hematological Dataset: Clinical Insights for Improved Diagnosis and Patient Management," v2, Mendeley Data, 2024. https://data.mendeley.com/datasets/6fsrsk3mb8/2

[21] D. Chicco and G. Jurman, "The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation," *BMC Genomics*, vol. 21, Art. 6, 2020. https://doi.org/10.1186/s12864-019-6413-7

[22] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. https://doi.org/10.1023/A:1010933404324

[23] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995. https://doi.org/10.1007/BF00994018

[24] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794. https://doi.org/10.1145/2939672.2939785

[25] W. J. Youden, "Index for rating diagnostic tests," *Cancer*, vol. 3, no. 1, pp. 32–35, 1950. https://doi.org/10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.0.CO;2-3

[26] A. J. Vickers and E. B. Elkin, "Decision curve analysis: A novel method for evaluating prediction models," *Med. Decis. Making*, vol. 26, no. 6, pp. 565–574, 2006. https://doi.org/10.1177/0272989X06295361

[27] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947. https://doi.org/10.1007/BF02295996

[28] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30*, 2017, pp. 4765–4774. https://doi.org/10.48550/arXiv.1705.07874

[29] G. S. Collins *et al.*, "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, e078378, 2024. https://doi.org/10.1136/bmj-2023-078378

[30] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011. https://jmlr.org/papers/v12/pedregosa11a.html

[31] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman & Hall, 1993.

[32] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[33] D-CHSS Dengue Predictor web application (Group 03, UAP), 2026. https://dengue-two-delta.vercel.app/

[34] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *Proc. 22nd Int. Conf. Machine Learning (ICML)*, 2005, pp. 625–632. https://doi.org/10.1145/1102351.1102430

[35] United Nations, "Goal 3: Ensure healthy lives and promote well-being for all at all ages," targets 3.3, 3.8 and 3.d. [Online]. Available: https://sdgs.un.org/goals/goal3

[36] World Health Organization, "Neglected tropical diseases," health topic page (list includes dengue and chikungunya). [Online]. Available: https://www.who.int/health-topics/neglected-tropical-diseases

[37] United Nations, "Goal 10: Reduce inequality within and among countries," targets 10.2 and 10.3. [Online]. Available: https://sdgs.un.org/goals/goal10

[38] United Nations, "Goal 9: Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation," targets 9.5 and 9.c. [Online]. Available: https://sdgs.un.org/goals/goal9

[39] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023. https://doi.org/10.1561/2200000101